F78AA Dec 2021 Final Exam Solutions

1. (a) $(1+i)(1-d_{12})^{12} = 1$

$i = \frac{1}{(1-0.0012)^{12}} - 1 = 0.014513$

(b) $(1+i_{12})^{12}(1-d_6)^6 = 1$

$d_6 = 1 - \frac{1}{(1+0.0021)^6} = 0.012508$

2. $167.50 = 10a_{\overline{5}|}^{0.092}$
$+ 10v^5[(1+0.01p)v + (1+0.01p)^2v^2 + (1+0.01p)^3v^3$
$+ \cdots]$

$16.75 = a_{\overline{5}|}^{0.092} + v^5 a_{\overline{\infty}|}^{j}$

where, $1 + j = \frac{1.092}{1+0.01p} \quad \Rightarrow \quad j = \frac{0.092-0.01p}{1+0.01p}$

$$\frac{16.75 - 3.869550}{0.644001} = \frac{1+0.01p}{0.092-0.01p}$$

$1 + 0.01p = 20.00067(0.092 - 0.01p)$

$p = 4.00$

3 (a) $2000 \times 1.04^{\frac{3}{12}} \times 1.025^{\frac{3}{12}}$
$= 2032.21$

(b) $2000 \times 1.04^{\frac{3}{12}} \times 1.025^{\frac{8}{12}} = 2053.2299 < 2200$

$2000 \times 1.04^{\frac{3}{12}} \times 1.025^{\frac{8}{12}} \times 1.03^{\frac{n}{12}} > 2200$

$1.03^{\frac{n}{12}} > 1.07148$

$n \geq 12 \times \frac{\ln 1.07148}{ln\,1.03} = 28.03$

$\therefore n = 29$

Least number of integer months = 3 + 8 + 29 = 40

4. (a) $P_A = 5a_{\overline{20}|}^{0.06} + 100v^{20@i=0.06}$

$P_A = 5(11.469921) + 100(0.311805)$

$P_A = 88.530105 \simeq 88.53$

(b) Check if CGT is payable.

(i) $i = 0.065 > \frac{5 \times 0.7}{100} = 0.035 \quad \Rightarrow \quad$ CGT is payable

$P_V = 0.7 \times 5a_{\overline{10}|}^{0.065} + [100 - 0.3(100 - P_V)]v^{10@i=0.065}$

$(1 - 0.3 \times 0.532726)P_V = 3.5(7.188830) + 70(0.532726)$

$P_V = \frac{62.451725}{0.840182} = 74.331187 \simeq 74.33$

(ii) $5a_{\overline{10}|}^{i} + 74.33v^{10} = 88.53$

Let, $f(i) = 5a_{\overline{10}|}^{i} + 74.33v^{10} - 88.53$
Initial Guess: $i < 0.06$
$f(0.05) = -4.2892 \quad < 0$
$f(0.04) = 2.2392 \quad > 0$
Linear Interpolation,

$$i^* = 0.04 + 0.01 \times \frac{0 - 2.2392}{-4.2892 - 2.2392} = 0.0434$$

5. (a) $\text{PV} = 6200a_{\overline{20}|}^{0.09} - 200(Ia)_{\overline{20}|}^{0.09}$

$\text{PV} = 6200(9.128546) - 200(70.905522)$

$\text{PV} = 42415.88$

(b) Capital outstanding immediately after the ninth repayment is made

$= 4400a_{\overline{11}|}^{0.09} - 200(Ia)_{\overline{11}|}^{0.09}$

$= 4400(6.805191) - 200(35.05329)$

$= 22932.1824$

Let, X = the tenth repayment under the new scheme

$(X + 200)a_{\overline{11}|}^{0.07} - 200(Ia)_{\overline{11}|}^{0.07} = 22932.1824$

$(X + 200)(7.498674) - 200(39.965154) = 22932.1824$

$X = 3924.09$

OR

$X\,a_{\overline{11}|}^{0.07} - 200(Ia)_{\overline{10}|}^{0.07}(1.07^{-1}) = 22932.1824$

$X(7.498674) - 200(34.739133)(1.07^{-1}) = 22932.1824$

$X = 3924.09$

6. (a) The value on 1 January 2030 of the repayments made in the calendar years 2030-2034

$$= 0.5Y\,a_{\overline{12}|}^{0.01}\left[1 + \left(\frac{1.05}{1.01^{12}}\right) + \left(\frac{1.05}{1.01^{12}}\right)^2 + \left(\frac{1.05}{1.01^{12}}\right)^3 + \left(\frac{1.05}{1.01^{12}}\right)^4\right]$$

$$= 0.5Y\,a_{\overline{12}|}^{0.01}\ \ddot{a}_{\overline{5}|}^{j}$$

$$\text{where, } 1 + j = \frac{1.01^{12}}{1.05} \quad \Rightarrow \quad j \approx 0.073167$$

$= 0.5Y(11.255077)(4.363134)$

$= 24.5537Y$

(b) The value on 1 January 2025 of the repayments made in the calendar years 2025-2029

$$= Y\, a_{\overline{4}|}^{0.04}\left[1+\left(\frac{1.1}{1.04^4}\right)+\left(\frac{1.1}{1.04^4}\right)^2+\left(\frac{1.1}{1.04^4}\right)^3+\left(\frac{1.1}{1.04^4}\right)^4\right]$$

$$= Y\, a_{\overline{4}|}^{0.01}\ \ddot{a}_{\overline{5}|}^{k}$$

where, $1+k=\frac{1.04^4}{1.1} \quad \Rightarrow \quad j \simeq 0.063508$

$$= Y\left(3.629895\right)\left(4.437452\right)$$

$$= 16.1075\,Y$$

(c) $\left[\left(24.5537\,Y\right)\left(1.04^{-20}\right)+16.1075\,Y\right]\times 1.05^{-5}=1000000$

$Y = 46727$